MATH2241 - Introduction to mathematical analysis

(1st semester 2026–2027)

Tutorial 1

1. Suppose that a, b are in an ordered field F and $a < b + \epsilon$ for any $\epsilon > 0$. Prove that $a \leq b$.

2. For each of the following sets, find the supremum and infimum (without proof) if they exist. State also whether the supremum and infimum are the maximum and minimum respectively.

   (a) $S_1 = \left\{x \in \mathbb{R} : x \neq 0, \frac{1}{x} > 2\right\}$

   (b) $S_2 = \left\{\frac{1}{k} + \frac{2}{m} + \frac{3}{n} : k, m, n \in \mathbb{N}\right\}$

   (c) $S_3 = \{3^x : x \in \mathbb{Z}\}$

3. Let A be a nonempty bounded subset of $\mathbb{R}$. What can we say about A if $\inf A = \sup A$?

4. Let A and B be nonempty bounded subsets of $\mathbb{R}$. Prove that if $A \subseteq B$, then
$$\inf B \leq \inf A \leq \sup A \leq \sup B$$

5. Let A and B be nonempty bounded subsets of $\mathbb{R}$. Define
$$A + B = \{a + b : a \in A \text{ and } b \in B\}$$
$$AB = \{a \times b : a \in A \text{ and } b \in B\}$$

   (a) Is that $\sup(A + B) = \sup A + \sup B$?

   (b) Is that $\sup(AB) = (\sup A) \times (\sup B)$?

6. Let A be a nonempty subset of $\mathbb{R}$ and let w be an upper bound of A. Suppose that for every $n \in \mathbb{N}$, there exists $a_n \in A$ such that $w - \frac{1}{n} \leq a_n$. Prove that $w = \sup A$.

7. (a) Give an example of a set of irrational numbers whose supremum is rational.

   (b) Give an example of a set of rational numbers whose supremum is irrational.

8. Show that

   (a) the root $\sqrt[m]{n}$ for $m, n \in \mathbb{N}$ is either an integer or an irrational number.

   (b) $\sqrt{n} + \sqrt{n+1}$ is an irrational number for every $n \in \mathbb{N}$.

   (c) $\sqrt{n + \sqrt{n}}$ is an irrational number for every $n \in \mathbb{N}$.